Week 9 Lower Bounds

Recall: in Gaussian sequence model,

$$Y = \theta + N, \quad N \sim \mathcal{N}(0, \sigma^2 I_d)$$

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E} \|\hat{\theta} - \theta\|_2^2 = \Theta(d\sigma^2)$$

↑ minimize over all $\hat{\theta} : y \to \mathbb{R}^d$

$$\inf_{\hat{\theta}} \sup_{\theta \in B_0(k)} \mathbb{E} \|\hat{\theta} - \theta\|_2^2 = \Theta(\sigma^2 k \log d)$$

↑ achieved by HT/ST estimators with threshold $\Theta(\sigma\sqrt{\log d})$

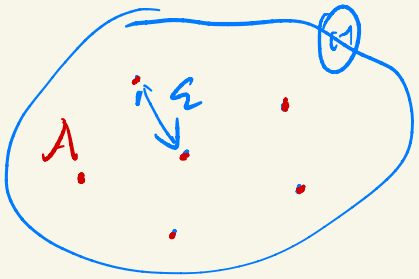
Three lower bounds methods (Bin Yu's 1997 paper):
- Le Cam's 2 point
- Assouad's
- Fano's inequality

From estimation to M-ary testing:

Definition: $\mathcal{A} \subseteq \Theta$ is called ε-packing, if $\forall \theta_1, \theta_2 \in \Theta$, $\theta_1 \neq \theta_2$, we have

$$d(\theta_1, \theta_2) > \varepsilon$$

(d: metric on Θ)



Estimation: $\forall\, \theta \in \Theta$

$Y \sim P_\theta$

Estimator: $\hat{\theta} : \mathcal{Y} \to \Theta$

Good estimator $\Rightarrow$ $d(\hat{\theta}, \theta)$ is small

$\Rightarrow$ good M-ary tester

Thm: suppose $\mathcal{A} = \{\theta_1 \cdots \theta_M\}$ is $2D$-packing ($D > 0$) and $\exists\ \hat{\theta} = \hat{\theta}(Y)$ such

$$\inf_{\theta \in \Theta} \mathbb{P}[d(\theta, \hat{\theta}) \leq D] \geq 1 - \delta \in [0,1]$$

Then $\exists\ \psi : \mathcal{Y} \to \{1 \cdots M\}$ s.t.

$$\min_{j \in \{1, \cdots, M\}} P_{\theta_j}[\psi(Y) = j] \geq 1 - \delta$$

Proof: $\psi(Y) = \underset{j = 1 \cdots M}{\operatorname{argmin}}\ d(\hat{\theta}(Y), \theta_j)$

· Fano's inequality

KL-divergence (relative entropy)

$$D(P\|Q) := \int \log \frac{dP}{dQ}(x)\, dP(x)$$

$\frac{dP}{dQ}$ → RN-derivative, ratio of densities

$$D(P\|Q) = -\int \log \frac{dQ}{dP}(x)\, dP(x)$$

$$\overset{\text{Jensen's}}{\geq} \mathbb{E}_{x\sim P}\left[-\log \frac{dQ}{dP}(x)\right]$$

$$\geq 0$$

· Mutual Information

$(W, Y) \sim P_{WY}$

$$I(W;Y) = D(P_{WY} \| P_W \times P_Y)$$

$$= \int D(P_{Y|W=w} \| P_Y)\, dP_W(w)$$

Fano's inequality

Thm (Cor 32 in Lec notes). Suppose $(W, Y) \sim P_{WY}$, W equiprobable on $\{1, \dots, M\}$, $\psi : \mathcal{Y} \to \{1 \cdots M\}$. Let $\delta := \mathbb{P}[\psi(Y) \neq W]$. Then

$$\inf_Q \frac{1}{M} \sum_{j=1}^{M} D(P_{Y|W=j} \| Q) \geq (1-\delta) \log M - h(\delta)$$

$= I(W;Y)$

binary entropy: $\delta \log \frac{1}{\delta} + (1-\delta) \log \frac{1}{1-\delta}$

- Construction of M-ary tests.

good M-ary test $\begin{cases} \text{well-separated under metric distance} \\ \text{close in that } D(P_\theta \| Q) \text{ is small} \end{cases}$

Example: If you pack exponentially points on

$$\{x : \|x\|_2 \leq \sqrt{d}\}$$

what is the minimum pairwise distance between these points?

(naively, I might pick the vertices of the unit cube $\{x : x_i = \pm 1\}$, with min distance is 1; however, there are packing achieving min distance $\sim \sqrt{d}$!)

Lem (Varshamov-Gilbert). $\exists\ C_1, C_2 > 0$, such that the following holds for integers k, d, $1 \le k \le d/8$:

$\exists\ w_1, \dots w_M \in \{0,1\}^d$, such that

- $\rho(w_i, w_j) \ge \frac{k}{2}, \quad i \ne j$

  Hamming distance $= \sum_{j=1}^{d} \mathbb{1}\, w_{il} \ne w_{jl}$

- $\log M \ge \frac{k}{8} \log\left(1 + \frac{d}{2k}\right)$

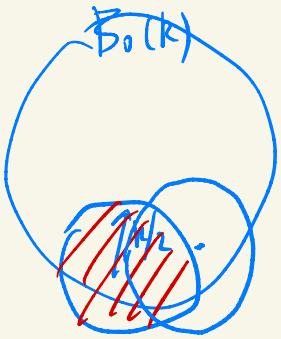
- $|w_j|_0 = k, \quad \forall j = 1 \dots M$

Proof sketch: choose $w_1 \dots w_M$ by the following:

For $i = 1 \dots M$, choose w_i as any element in

$$B_0(k) \setminus \bigcup_{j=1}^{i-1} B(w_j, k/2) \qquad (*)$$

$B(w_j, k/2)$: Hamming Ball of radius $k/2$, centered w_j



Need to show $(*)$ is nonempty for $i = M$, which is true if

$$\exp\left(d\, h\left(\tfrac{k}{d}\right) + d(\log d)\right) \ge M \cdot \exp\left(d\, h\left(\tfrac{k}{2d}\right) + o(\log d)\right)$$

which is true if

$$M \le \exp\left(k \log \frac{d}{k} + O(k \log(d))\right)$$

$h(p) \approx p \log \frac{1}{p}$ when p is small

■

- Proof of lower bound for $\Theta = B_0(k)$

Choose $\{w_1, \cdots, w_M\}$ as in VG lemma,

$$\|w_i - w_j\|_2^2 \ge \frac{k}{2}$$

$$\log M \ge \frac{k}{8} \log\left(1 + \frac{d}{2k}\right)$$

Define

$$\theta_j = \beta w_j \quad j = 1, \cdots, M$$

($\beta \in (0, \infty)$ to be chosen later)

Choose $Q = N(0, \sigma^2 I_d)$

$$\frac{1}{M} \sum_{j=1}^{M} D(P_{\theta_j} \| Q) = D(P_{\theta_1} \| P_0)$$

$$= \frac{k\beta^2}{2} \log e$$

Experience tells us: choose θ_j such that δ in Fano's inequality to be constant order (if my goal is to lower bound the expected risk)

Fano's inequality $\Rightarrow$ any test χ satisfies

$$\mathbb{P}[\chi(Y) = W] \log M - \left| \le \frac{k\beta^2}{2} \log e \right.$$

$$\mathbb{P}[\chi(Y) = W] \le O(d) + \frac{\frac{k\beta^2}{2} \log e}{\frac{k}{8} \log\left(1 + \frac{d}{2k}\right)}$$

Choose $\beta = o\left(\sqrt{\log \frac{d}{k}}\right)$ can make $\mathbb{P}[\chi(\eta) = \omega] < \frac{1}{2}$

$\Rightarrow$ estimation error is at least

$$\|\theta_i - \theta_j\|_2 \sim \sigma\beta\sqrt{k}$$

$$\sim \sigma\sqrt{k \log \frac{d}{k}}$$

$$\sim \begin{cases} \sigma\sqrt{k \log d} & (\text{if } k = d^{1-\varepsilon},\ \varepsilon > 0) \\ \sigma\sqrt{d} & (\text{if } k \sim d) \end{cases}$$

Thm: in Gaussian sequence model,

$$\inf_{\hat{\theta}} \sup_{\theta \in B_0(k)} \mathbb{E}\|\hat{\theta} - \theta\|_2^2 \gtrsim \sigma^2 k \log d/k$$

Empirical Risk Minimization & Metric Entropy

$$B_q(R) = \{\theta \in \mathbb{R}^d : \sum_{i=1}^d |\theta_i|^q \le R^q\}$$

smaller than <u>weak l_q-norm ball</u>.

order $|\theta_1^*| \ge |\theta_2^*| \cdots$

$$|\theta_i^*| \le \frac{R}{i^{1/q}}$$

suppose $Y_1 \cdots Y_n$ iid $\sim N(\theta^*, \sigma^2 I)$, $\theta^* \in B_q(R)$, $q \in (0,1]$

ERM: $\hat{\theta} = \underset{\theta \in B_q(R)}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n \|Y_i - \theta\|_2^2$

By optimality,

$$\frac{1}{n}\sum_{i=1}^n \|Y_i - \hat{\theta}\|_2^2 \le \frac{1}{n}\sum_{i=1}^n \|Y_i - \theta^*\|_2^2$$

$w_i + (\theta^* - \hat{\theta})$ $\qquad$ $Y_i = \theta^* + w_i$

$$\Rightarrow \quad \|\theta^* - \hat{\theta}\|^2 \le \frac{2}{n}\sum_{i=1}^n \langle w_i, \hat{\theta} - \theta^* \rangle$$

$$= \frac{2}{\sqrt{n}} \langle \sum_{i=1}^n \frac{w_i}{\sqrt{n}}, \hat{\theta} - \theta^* \rangle \qquad (*)$$

If no l_q constraint,

$$\Rightarrow \mathbb{E}\|\theta^* - \hat{\theta}\|^2 \le \frac{2}{\sqrt{n}} \sqrt{d\sigma^2} \cdot \sqrt{\mathbb{E}\|\hat{\theta} - \theta^*\|^2}$$

$$\Rightarrow \text{MSE} \le \frac{4d\sigma^2}{n}$$

want: upper bound $\mathbb{E}[\langle \sum_{i=1}^n \frac{w_i}{\sqrt{n}}, \hat{\theta} - \theta^* \rangle]$ in terms of MSE

Define $P_z = \frac{1}{d}\sum_{i=1}^{d} P_{\hat{\theta}_i - \theta_i^*}$, $\quad \Sigma := \mathbb{E}\left[\frac{1}{d}\sum_{i=1}^{d} |\hat{\theta}_i - \theta_i^*|^2\right]$

$\Rightarrow \mathbb{E}|z|^2 = \Sigma$, $\quad \mathbb{E}|z|^q \le \frac{2R^q}{d}$

Lem: suppose $G \sim N(0,1)$, $\mathbb{E}|z|^2 = \Sigma$, $\mathbb{E}|z|^q = a$

Then $\quad \mathbb{E}[zG] \le \sqrt{2e\ln(\Sigma a^{-2/q}) + 1} \cdot \Sigma^{\frac{1-q}{2-q}} a^{\frac{1}{2-q}}$

Intuition: $\mathbb{E}|z| \le (\mathbb{E}|z|^2)^{\frac{\alpha}{2}} (\mathbb{E}|z|^q)^{\frac{1-\alpha}{q}}$

where $\frac{\alpha}{2} + \frac{1-\alpha}{q} = 1$

$\alpha \in (0,1)$ if $q < 1$

Thm: ERM achieves $\mathbb{E}\|\hat{\theta} - \theta^*\|^2 \lesssim R^q \left(\sigma\sqrt{\frac{\log d}{n}}\right)^{2-q}$

Pf: $\frac{1}{\sqrt{n}}\mathbb{E}\left[\left\langle \sum_{i=1}^{n} \frac{w_i}{\sqrt{n}}, \hat{\theta} - \theta^* \right\rangle\right] \le \frac{d\sigma}{\sqrt{n}} \sup \mathbb{E}[zG]$

coupling of z and $G \sim N(0,1)$

$\overset{\text{Lem}}{\lesssim} \frac{d\sigma}{\sqrt{n}} \sqrt{\ln \cdots} \cdot \Sigma^{\frac{1-q}{2-q}} R^{\frac{q}{2-q}} d^{\frac{-1}{2-q}}$

from $(*)$

$\Rightarrow d\Sigma \lesssim \frac{d\sigma}{\sqrt{n}} \sqrt{\ln \cdots} \cdot \Sigma^{\frac{1-q}{2-q}} R^{\frac{q}{2-q}} d^{\frac{-1}{2-q}}$

$\Rightarrow \Sigma d \lesssim R^q \left(\sigma\sqrt{\frac{\log d}{n}}\right)^{2-q}$

Extension : $Y = X\theta^* + w$

Ref : Raskutti, Wainwright, Yu 2011

suppose $\frac{1}{\sqrt{n}} \max_{j=1,\cdots,d} \|X_j\|_2 \le 1$

$w \sim N(0, \sigma^2 I_{n\times n})$

$\frac{d}{R n^{\frac{q}{2}}} \gtrsim d^{\delta} \gtrsim c$, for some $\delta > 0$ small

$q \in (0, 1]$

$\hat{\theta} = \underset{\theta \in B_q(R)}{\operatorname{argmin}} \|Y - X\theta\|^2$

(note R^q here $= R_q$ in RWY 2011)

Thm 4 in RWY 2011 :

$$\frac{1}{n}\mathbb{E}\left[\|X\hat{\theta} - X\theta^*\|^2\right] \lesssim R^q \left(\frac{\sigma^2 \log d}{n}\right)^{1-\frac{q}{2}}$$

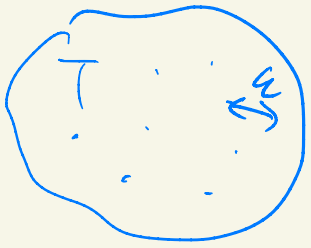
pf : Optimality of $\hat{\theta}$

$$\Rightarrow \frac{1}{2n}\|X(\hat{\theta} - \theta^*)\|^2 \le \frac{1}{\sqrt{n}} \left\langle w, \frac{1}{\sqrt{n}} X(\hat{\theta} - \theta^*) \right\rangle \quad (*)$$

· Dudley integral:

$$\mathbb{E}\left[\sup_{t\in T} X_t\right] \le 24 \int_0^\infty \sqrt{\log N(T,\varepsilon)}\, d\varepsilon$$

$N(T,\varepsilon)$: covering number under $d(t_1,t_2) = \|X_{t_1} - X_{t_2}\|_{\psi_2}$



chaining

$$X_t = X_{\pi_0(t)} + \left(X_{\pi_1(t)} - X_{\pi_0(t)}\right) + \left(X_{\pi_2(t)} - X_{\pi_1(t)}\right) + \cdots$$

↑ projection onto optimal $2^{-i}D$-packing

D = diameter

suppose $\delta^2 = \mathbb{E}\left\|\frac{X}{\sqrt{n}}(\hat{\theta} - \theta^*)\right\|^2$.

let's assume concentration, so that

$$\left\|\frac{X}{\sqrt{n}}(\hat{\theta} - \theta^*)\right\|_2 \lesssim \delta \quad \text{almost surely}$$

(Details using the "slicing" argument)

We will see (Appendix C in RWY paper) that

$$\log N\left(\frac{X}{\sqrt{n}} B_q(2^{\frac{1}{q}} R), \varepsilon\right) \asymp \left(\frac{R}{\varepsilon}\right)^{\frac{2q}{2-q}} \log d \quad (\#)$$

Then

$$\langle w, \tfrac{1}{\sqrt{n}} X(\hat{\theta} - \theta^*) \rangle \lesssim \sigma \int_0^{\delta} \sqrt{\log N\left(\tfrac{X}{\sqrt{n}} B_q(2^{\frac{1}{q}} R), \varepsilon\right)}\, d\varepsilon$$

$$\lesssim \sigma \int_0^{\delta} \left(\frac{R}{\varepsilon}\right)^{\frac{q}{2-q}} \sqrt{\log d}\, d\varepsilon$$

$$\asymp \sigma R^{\frac{q}{2-q}} \sqrt{\log d}\, \delta^{1-\frac{q}{2-q}}$$

back to $(*)$

$$\Rightarrow \quad \delta^2 \lesssim \frac{\sigma}{\sqrt{n}} R^{\frac{q}{2-q}} \sqrt{\log d} \cdot \delta^{\frac{2-2q}{2-q}}$$

$$\Rightarrow \quad \delta^2 \lesssim \text{bound in the theorem}$$

- Intuition behind $(\#)$ based on rate-distortion theory

$$R_Z(\delta) = \min I(u; z)$$

$$P_{uz} : \mathbb{E}|u - z|^2 \le \delta$$

Suppose $z \in \mathbb{R}$, $\mathbb{E}|z|^q \le \frac{R^q}{d}$

$$R_z(z) \asymp H(\text{Bern}(p))$$

$$\asymp p \log \frac{1}{p}$$

define p by $\begin{cases} \lambda^q p = \frac{R^q}{d} \\ \lambda^2 p = \delta^2 \end{cases}$

$$\Rightarrow p = R^{\frac{2q}{2-q}} d^{-\frac{2}{2-q}} \delta^{-\frac{2q}{2-q}}$$

$$R_{\vec{z}}(\delta\sqrt{d}) \asymp dp \log \frac{1}{p}$$

(choose $\varepsilon \leftarrow \delta\sqrt{d}$)

$$\lesssim R^{\frac{2q}{2-q}} \varepsilon^{-\frac{2q}{2-q}} \frac{\log \frac{1}{p}}{\log d}$$